

Achievement 2.3: Complex ML Models and Keras Part 2

Overview

This report presents the development and evaluation of a machine learning model designed to identify the most important indicators of a “pleasant day” using historical European weather data for the nonprofit organization ClimateWins. Using past weather observations from multiple weather stations across Europe, a Random Forest classification model was applied to determine which weather features — such as temperature, precipitation, and solar conditions — are most influential in predicting pleasant versus unpleasant days.

The analysis was conducted in two main stages:

1. All-Stations Model

First, a comprehensive model was built using data from all 15 weather stations and 135 weather-related features. This model was used to assess overall predictive performance and to identify which weather stations contributed most strongly to the classification of pleasant days.

2. City-Specific Models

Based on the all-stations feature importance results, separate Random Forest models were developed for the three most influential stations — **Budapest, Ljubljana, and Madrid**. These city-specific models included all available years of data and focused on understanding which local weather variables were most important for pleasant-day predictions at each location.

Each model’s performance was evaluated using test-set accuracy, and feature importance visualizations were created to support interpretation of the results. The findings reveal clear differences in how weather conditions drive pleasant-day outcomes across different European climates, with precipitation and temperature emerging as dominant predictors depending on location.

The results of this analysis can help ClimateWins prioritize data collection, improve forecasting strategies, and better understand regional climate patterns that influence day-to-day comfort. In addition, this project demonstrates practical machine learning skills — including data preparation, model training, evaluation, and interpretability — that are essential for applying data science techniques to real-world climate and sustainability challenges.

Methods Summary

1. Data Preparation

The analysis began with a cleaned and prepared European weather dataset provided for the ClimateWins project. The final dataset included observations from 15 weather stations, each reporting 9 standardized weather variables (e.g., precipitation, temperature, sunshine, cloud cover, pressure, humidity). After preprocessing in earlier exercises, the resulting feature matrix contained 135 total weather features (15 stations × 9 variables) and 22,950 daily observations, with no missing values. A corresponding target dataset was used to label each day as either pleasant (1) or unpleasant (0) based on predefined criteria.

2. Time Handling

Unlike the example notebook, this analysis did not explicitly filter the data by decade. Date variables were removed during preprocessing, meaning the models were trained on all available years of historical data rather than a fixed time window. As a result, findings should be interpreted as representative of the full dataset rather than a single decade.

3. Feature Selection and Reshaping

All 135 weather features were retained for the initial all-stations model to allow the Random Forest algorithm to determine which stations and variables were most predictive. Date-related fields were excluded, as the focus was on meteorological conditions rather than temporal patterns.

For the city-specific models, the feature set was reduced to 9 weather variables from a single station, allowing a focused examination of local drivers of pleasant weather outcomes.

4. Modeling with Random Forest

A Random Forest Classifier was selected due to its strong performance on structured tabular data and its ability to provide interpretable feature importance scores. Each model used 100 decision trees and a fixed random state for reproducibility.

The data was split into training (75%) and testing (25%) sets. Model performance was evaluated using test-set accuracy to assess predictive reliability.

5. Feature Importance Analysis

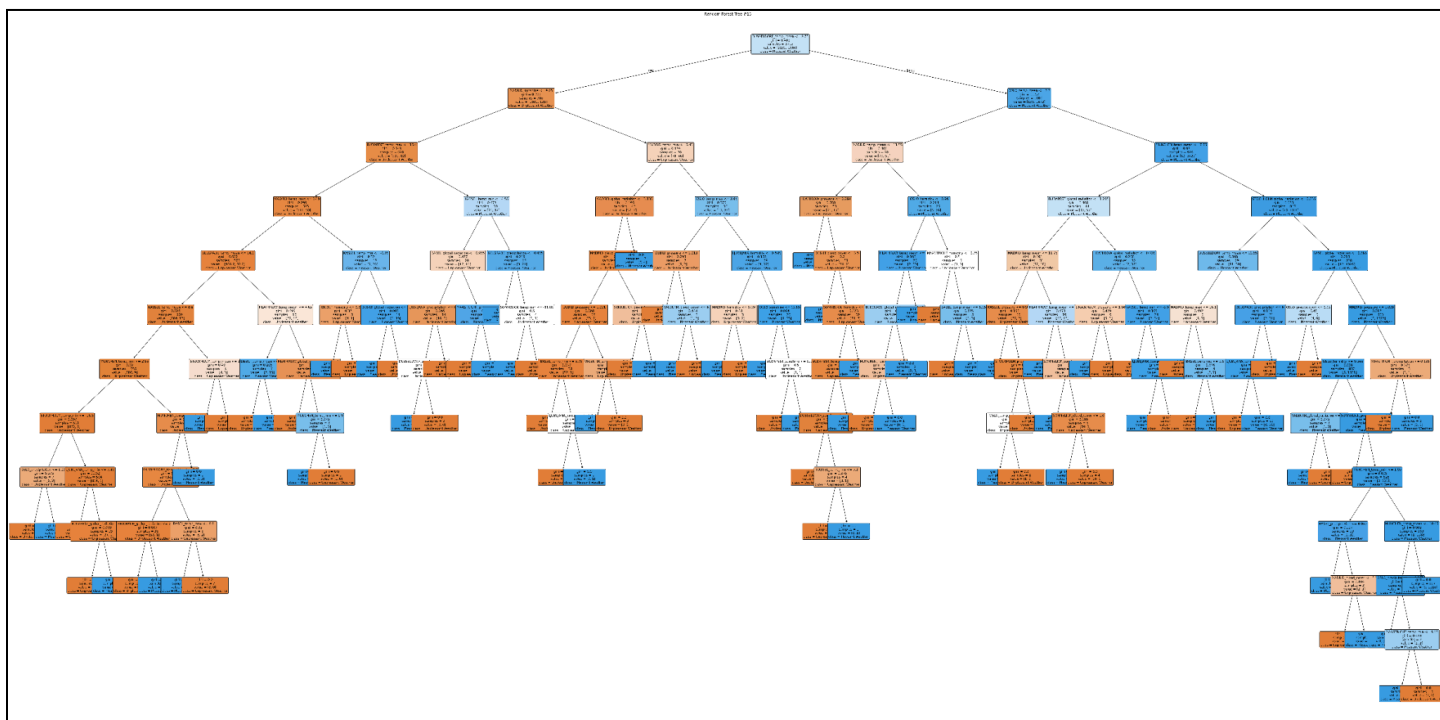
After training, feature importance values were extracted from each model to identify which weather variables contributed most to predictions.

- In the all-stations model, feature importances were reshaped and aggregated by station to determine which locations had the strongest overall influence.
- This analysis identified Budapest, Ljubljana, and Madrid as the three most influential stations.

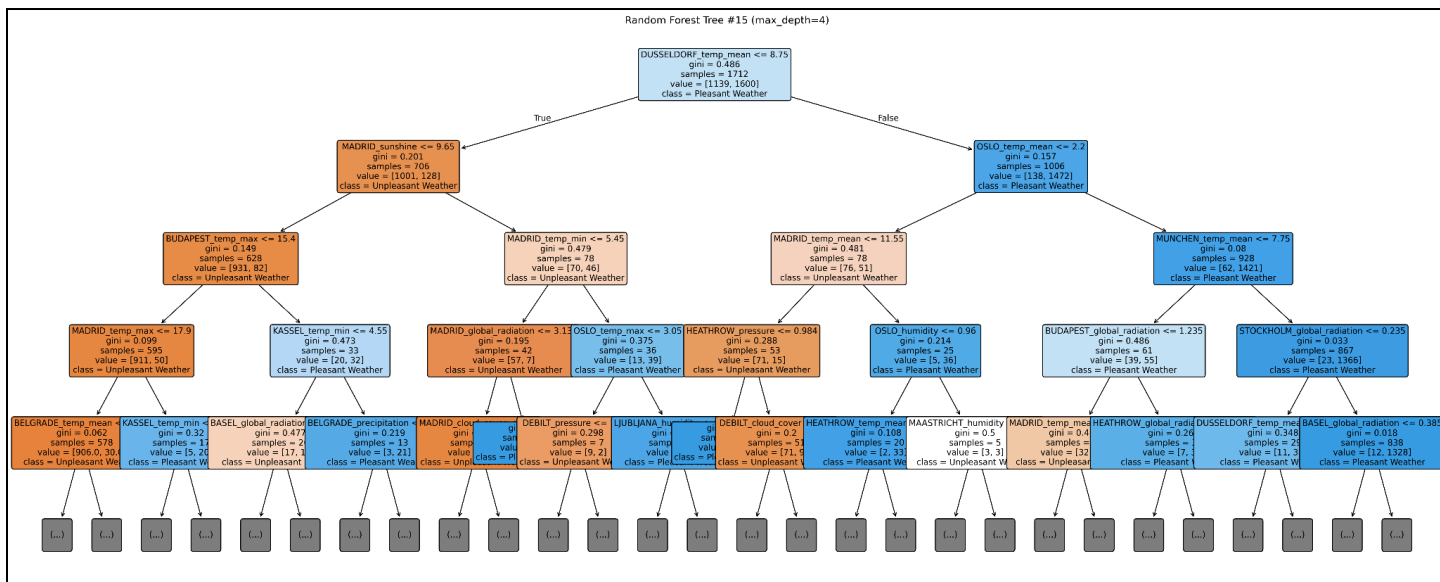
Separate city-level Random Forest models were then built for these stations, and feature importance plots were generated to compare how local weather drivers differed across locations.

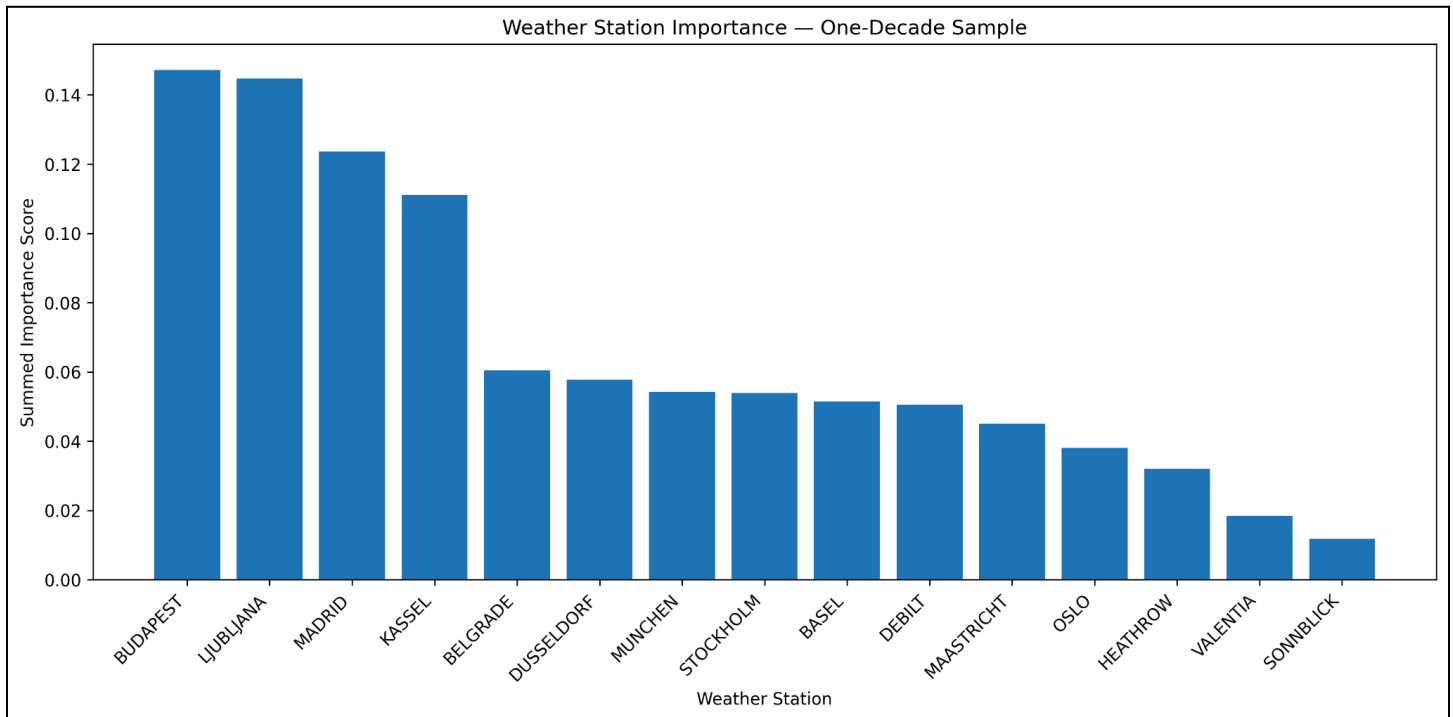
Random Forest Results: All Station, One Decade Sample

Full Tree



Tree with 4 layers of depth



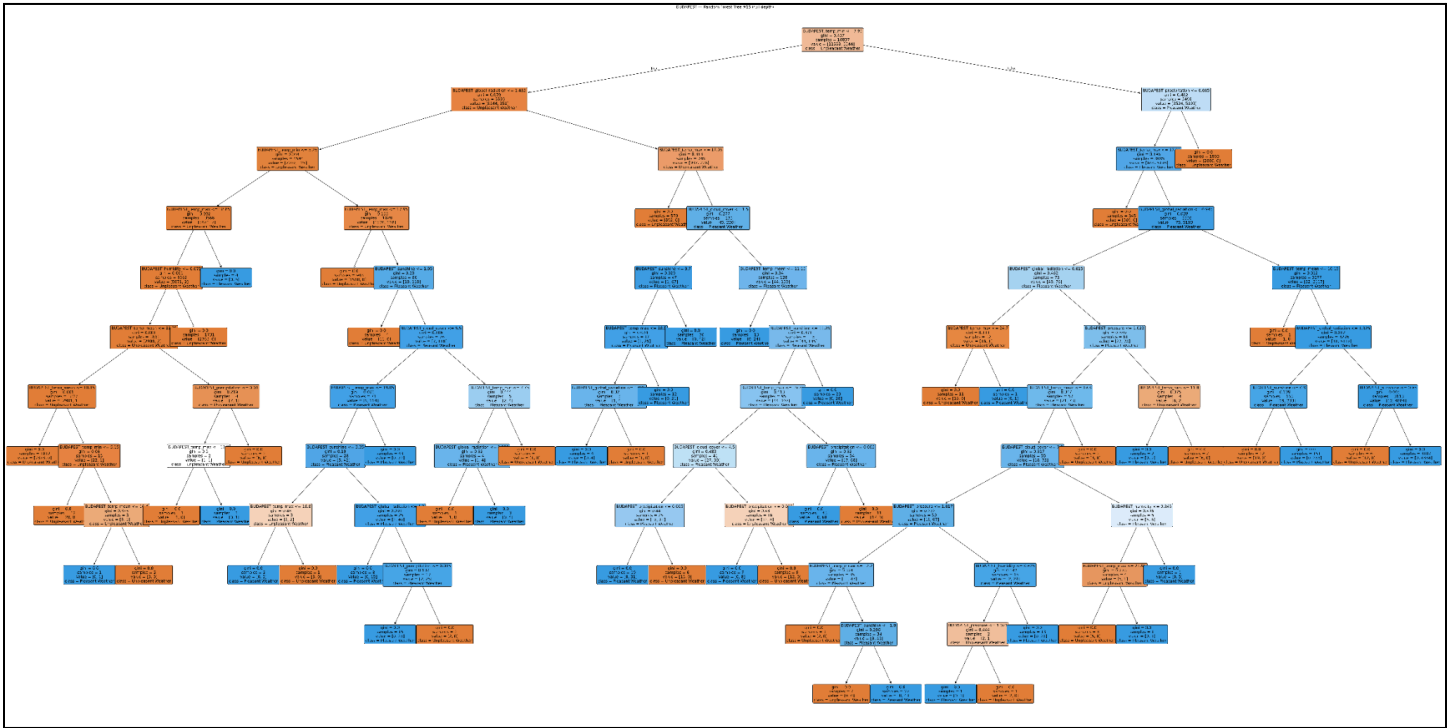


Observation & Insights

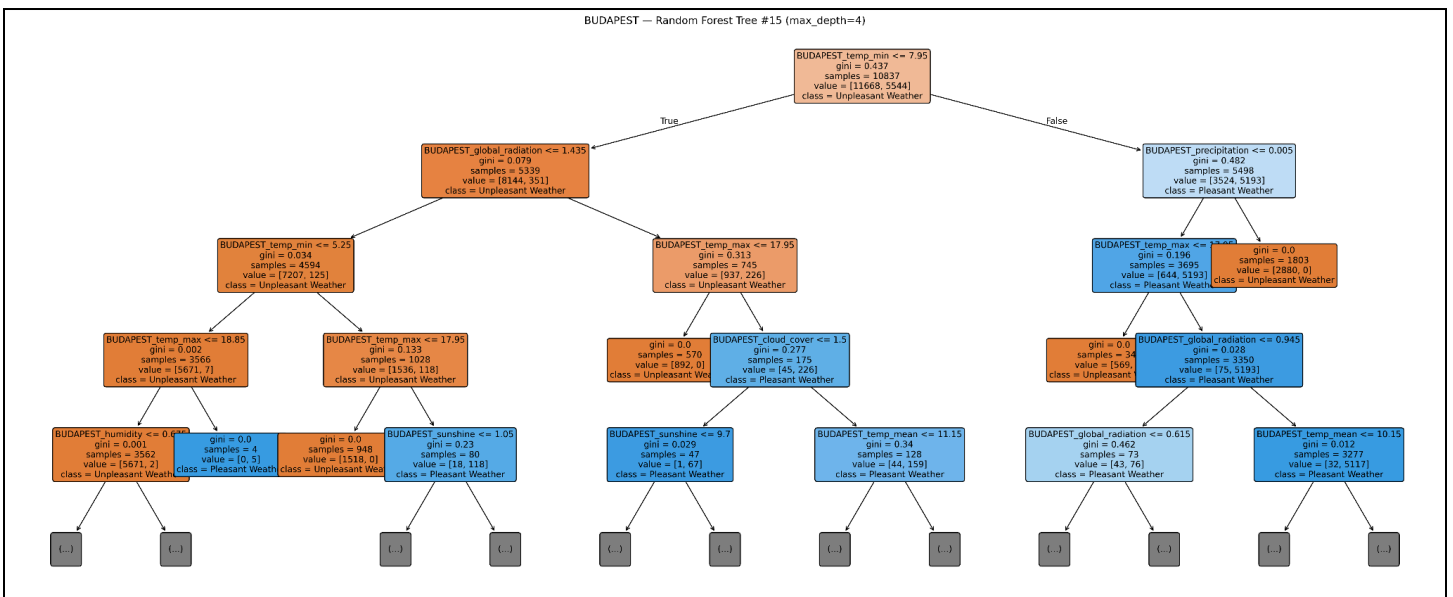
- The Random Forest Model achieved a high test accuracy of 97.26%.
- The model incorporated 135 features drawn from 15 weather stations, with 9 observation types per station (e.g., temperature, humidity, pressure, wind, sunshine). Feature importances were extracted from the trained model and reshaped to evaluate the relative contribution of each station.
- This analysis used a one-decade sample from the dataset. Under this formulation, **Budapest, Ljubljana, and Madrid** emerged as the most influential stations, collectively accounting for a substantial share of the total feature importance.
- Stations such as Kassel, Belgrade, Düsseldorf, and Munich contributed moderate importance, suggesting that their environmental signals still provide useful predictive information.
- The results suggest that predictive leverage is unevenly distributed across the station network. Rather than one centrally dominant station, a small group of stations provides disproportionate predictive value for identifying pleasant weather across regions. This insight can inform ClimateWins' decisions on sensor prioritization, data weighting in future models, and strategic investment in stations that offer early or broadly informative climate signals.

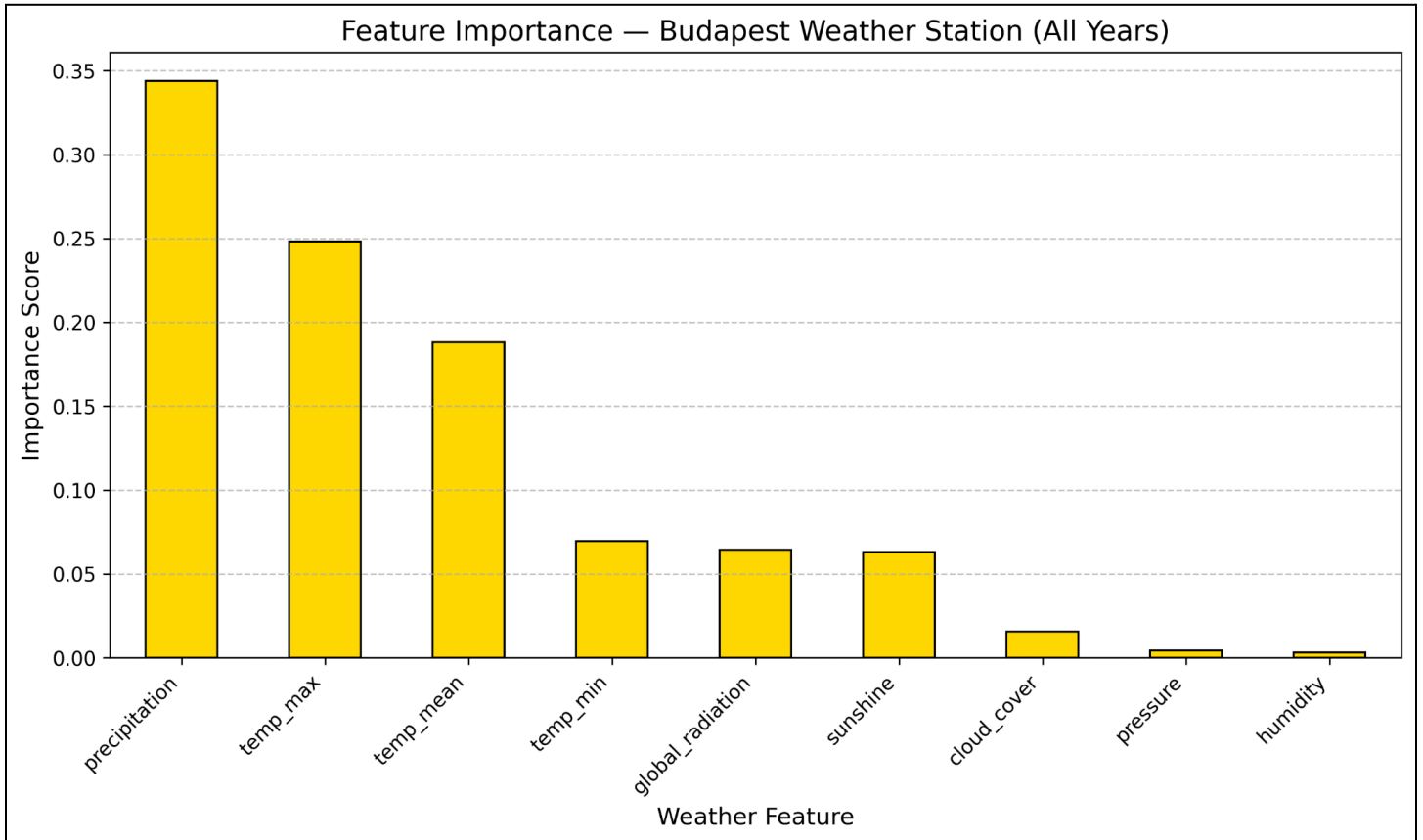
Random Forest Results: Budapest, All Years

Full Tree



Tree with 4 layers of depth



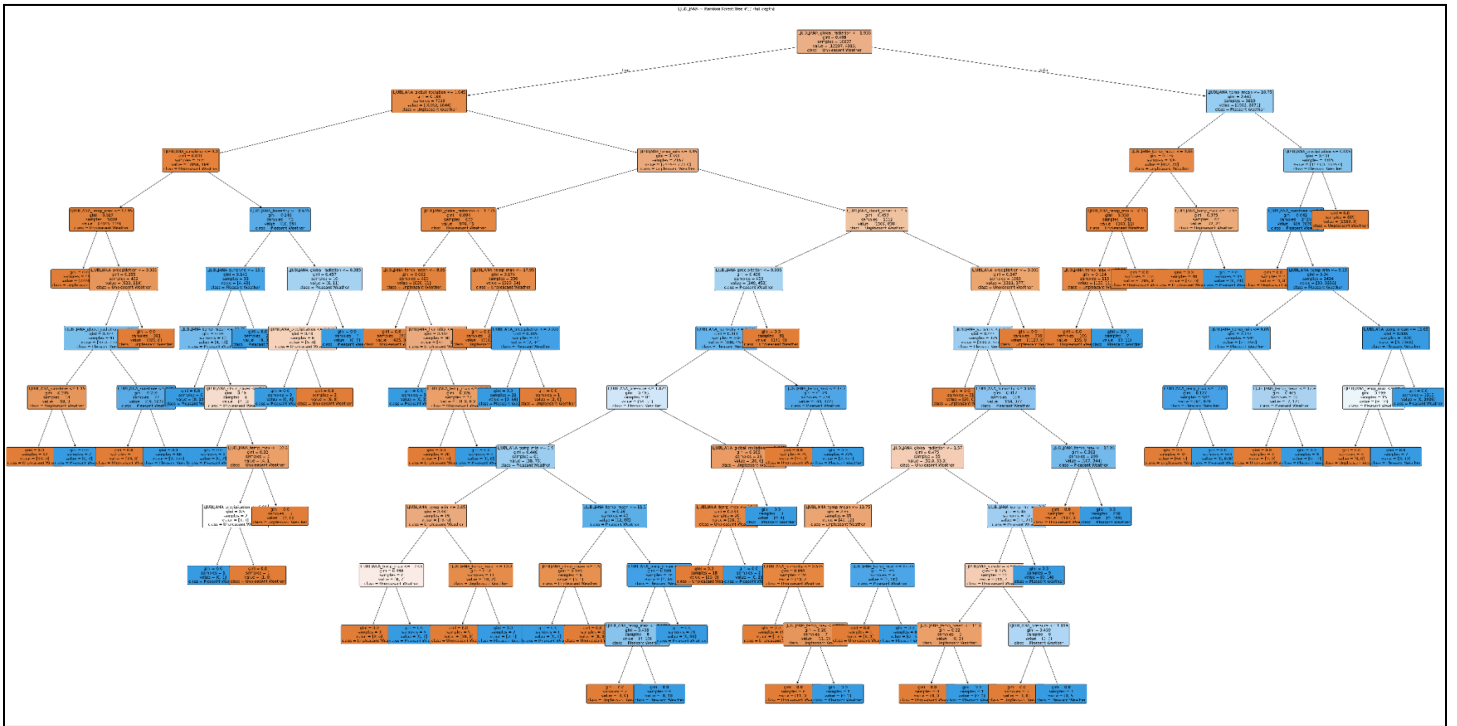


Budapest Random Forest — Key Insights

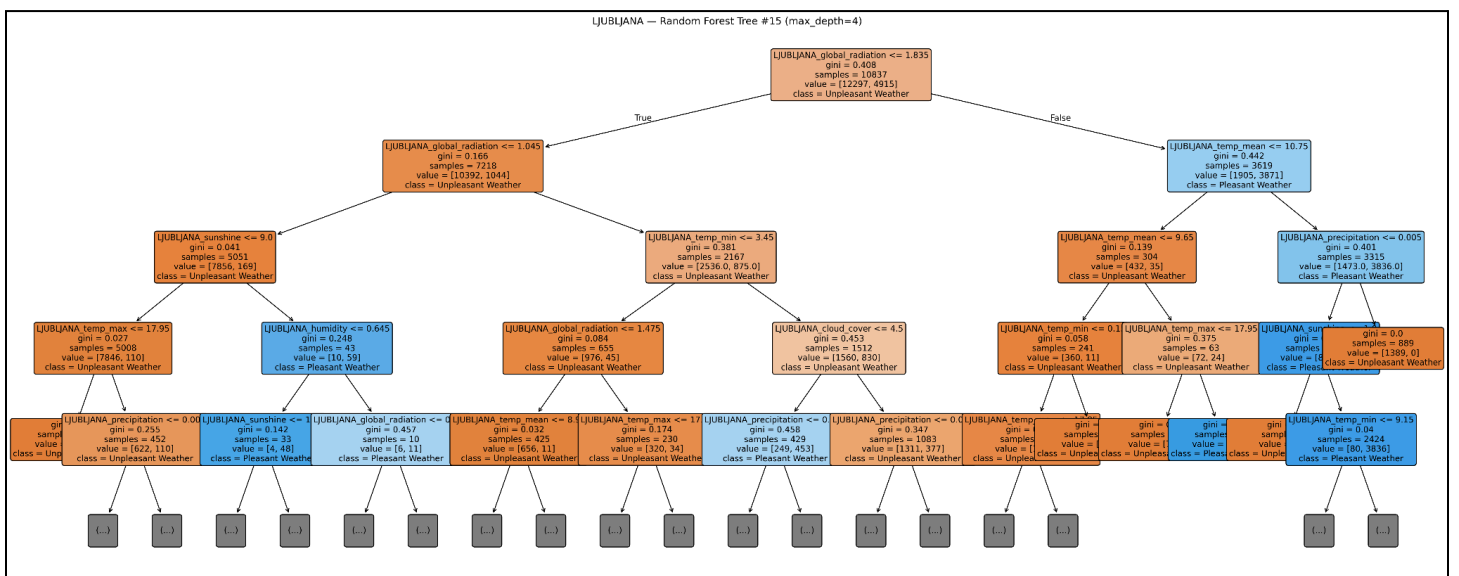
- The Random Forest model achieved 100% accuracy on the test set, indicating that Budapest's local weather indicators are highly predictive of pleasant days in this dataset.
- Precipitation is the most influential feature by a wide margin, suggesting that the presence or absence of rainfall is the primary driver of whether a day is perceived as pleasant in Budapest.
- Maximum and mean temperature are the next most important features, highlighting the role of warmth in shaping pleasant weather conditions.
- Global radiation and sunshine contribute meaningfully but less strongly, indicating that light conditions refine predictions once precipitation and temperature are accounted for.
- Pressure and humidity have minimal importance, suggesting they add little incremental predictive value for pleasant-day classification in Budapest.
- For Budapest, ClimateWins could prioritize high-quality precipitation and temperature monitoring over additional investment in pressure or humidity sensors when optimizing pleasant-weather prediction models.

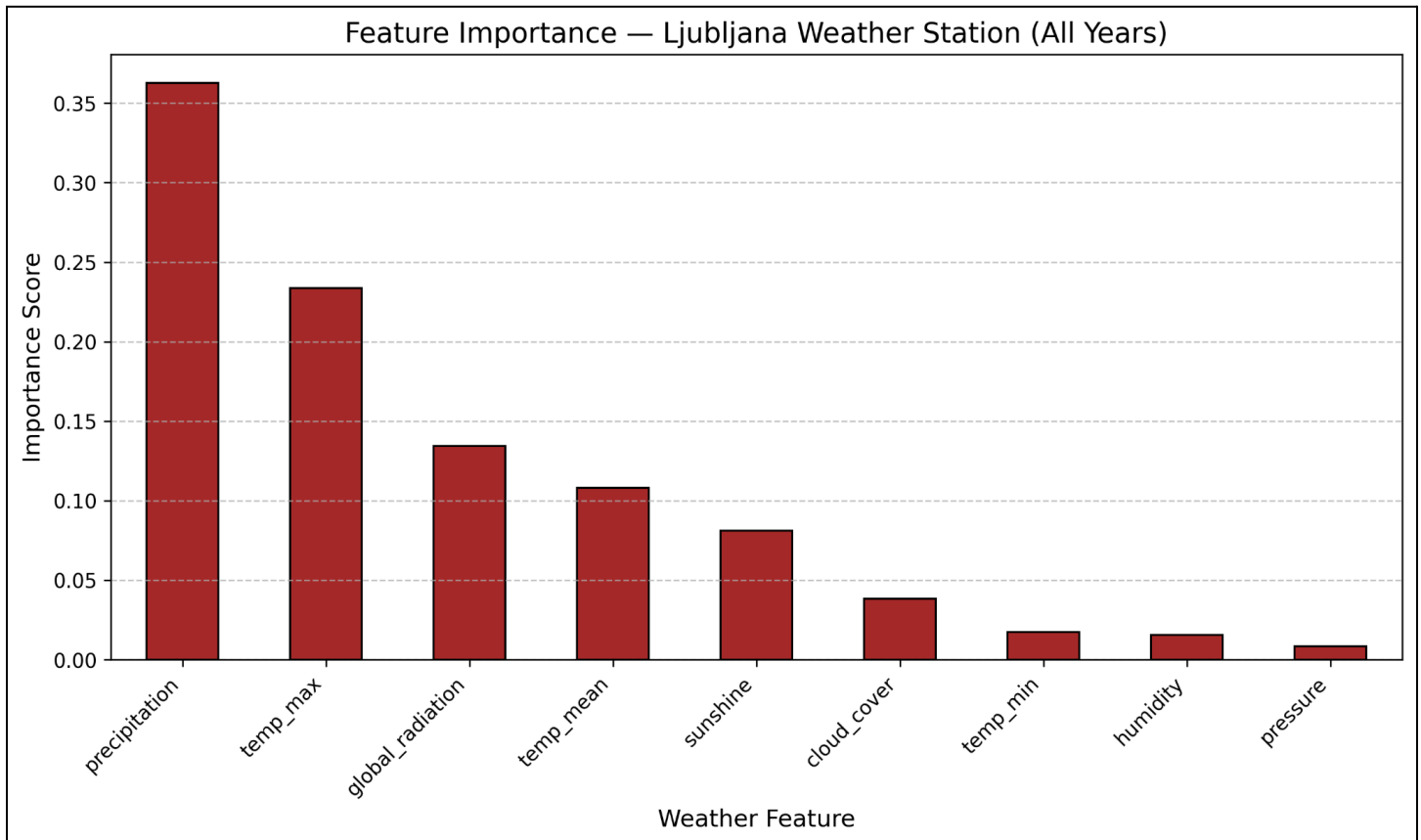
Random Forest Results: Ljubljana, All Years

Full Tree



Tree with 4 layers of depth



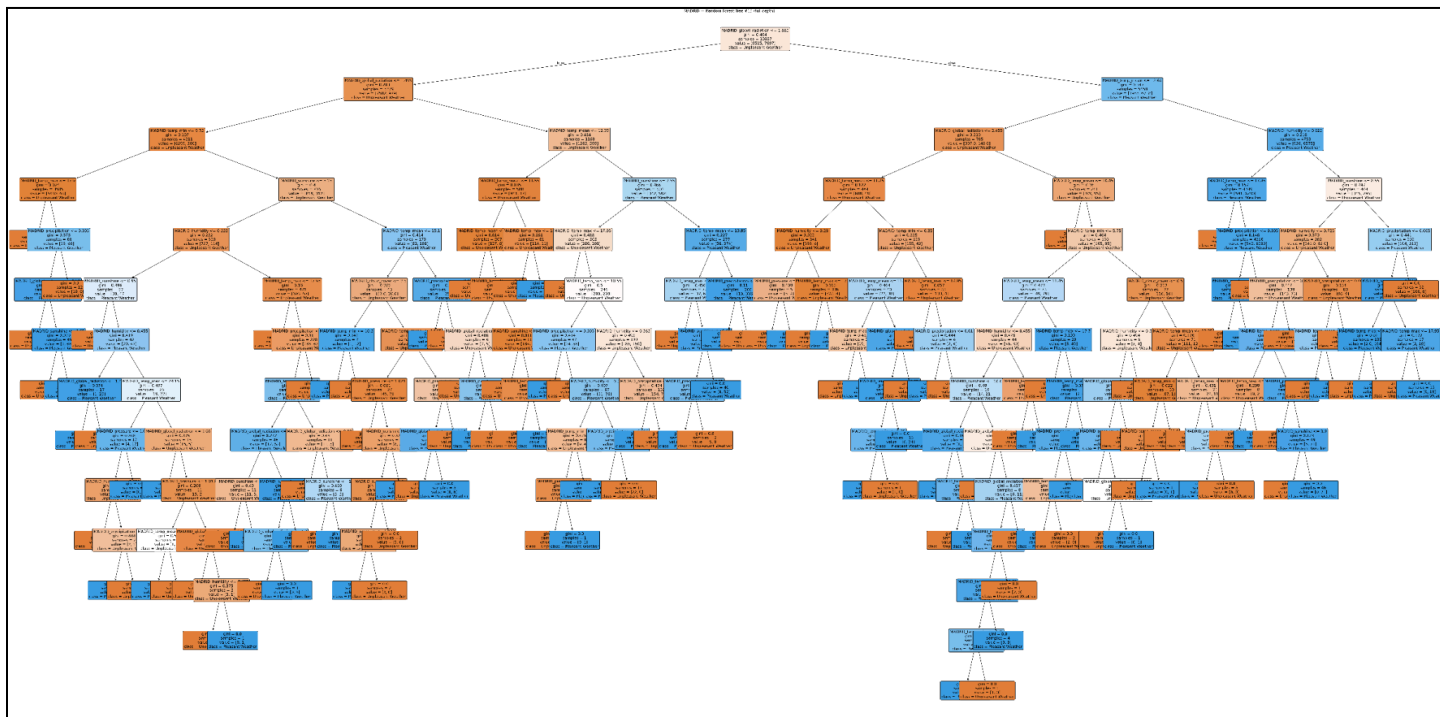


Ljubljana Random Forest - Key Insights

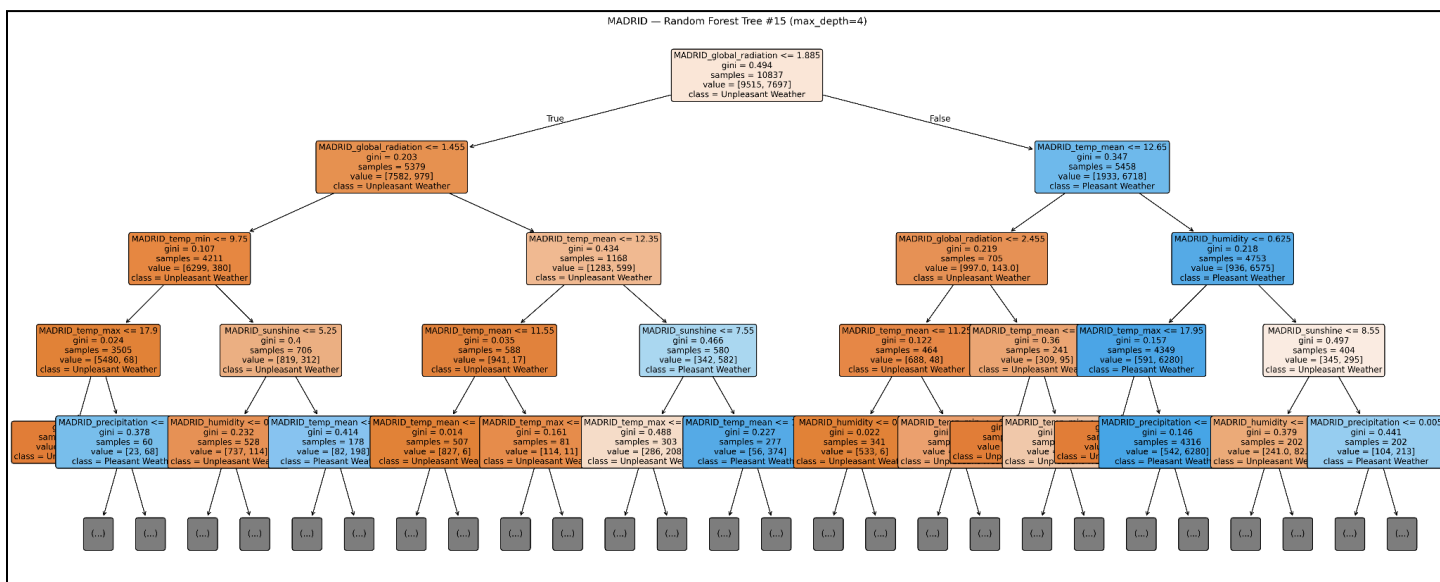
- The Random Forest model achieved 100% accuracy on the test set, indicating that Ljubljana's local weather indicators provide a very strong and internally consistent signal for predicting pleasant days in this dataset.
- Precipitation is the most influential feature by a substantial margin, reinforcing the importance of rainfall conditions in determining whether a day is classified as pleasant in Ljubljana.
- Maximum temperature and global radiation emerge as key secondary drivers, suggesting that both warmth and solar energy meaningfully shape pleasant weather conditions.
- Mean temperature and sunshine provide additional predictive value, refining the model's decisions once precipitation and peak temperature effects are accounted for.
- Cloud cover, minimum temperature, humidity, and pressure contribute relatively little to the model, indicating that these variables add minimal incremental information for pleasant-day classification in Ljubljana.
- For Ljubljana, ClimateWins could prioritize precipitation, temperature extremes, and solar radiation measurements when designing or optimizing models for pleasant-weather prediction, with lower priority placed on pressure and humidity monitoring.

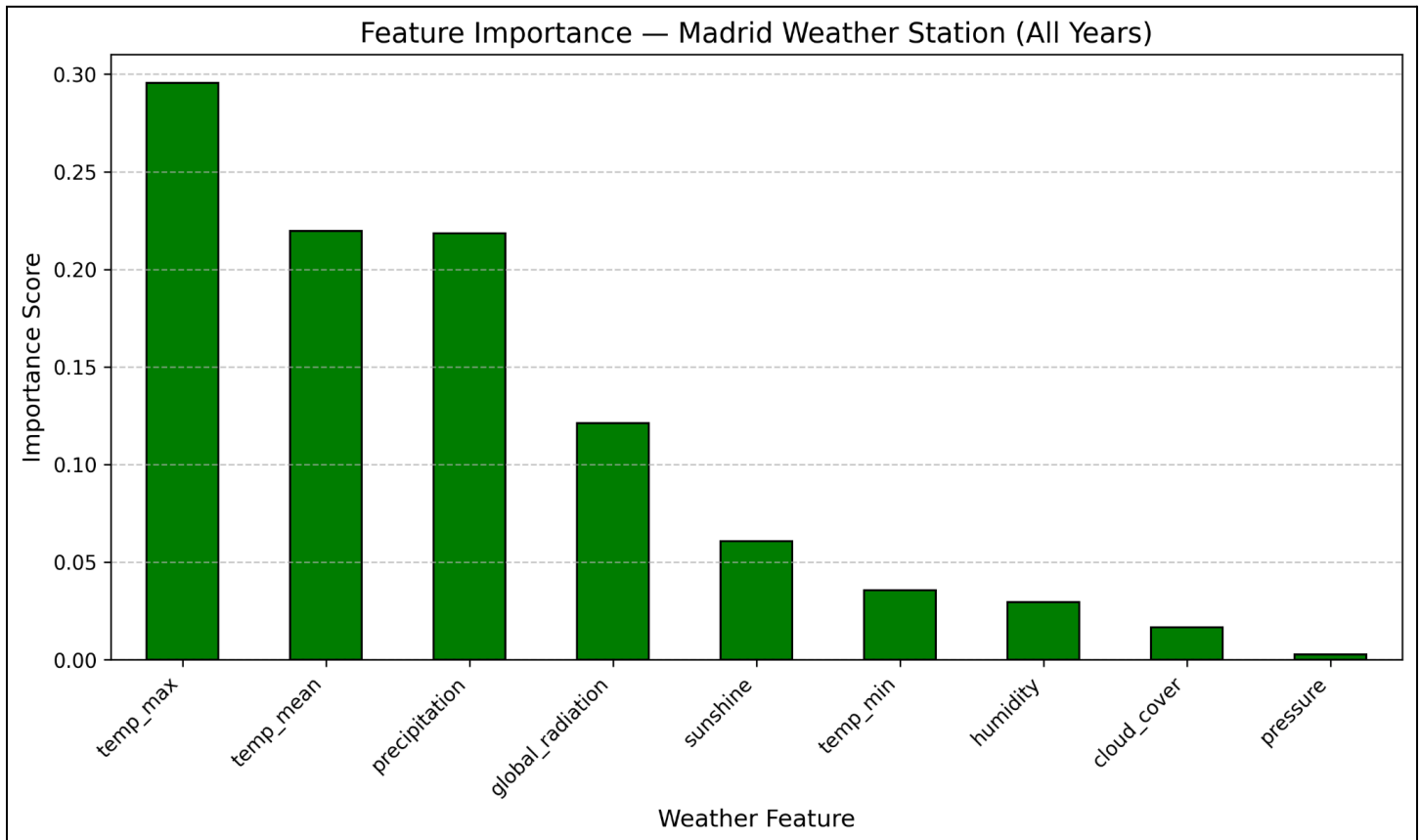
Random Forest Results: Madrid, All Years

Full Tree



Tree with 4 layers of depth





Madrid Random Forest — Key Insights

- The Random Forest model achieved 99.98% accuracy on the test set, indicating that Madrid's weather variables provide an exceptionally strong signal for classifying pleasant versus unpleasant days.
- Maximum temperature is the single most influential feature, with mean temperature close behind. This reflects Madrid's climate profile, where thermal comfort is the primary determinant of perceived pleasantness.
- Precipitation ranks among the top features, confirming that even in a relatively dry climate, rainfall strongly disrupts pleasant weather conditions.
- Global radiation and sunshine contribute meaningful additional explanatory power, highlighting the role of solar exposure in shaping comfort once temperature thresholds are met.
- Minimum temperature has moderate importance, suggesting overnight or early-morning conditions matter, but less than daytime heat in Madrid's context.
- Pressure, cloud cover, and humidity contribute very little to the model, indicating that these variables add limited incremental value for pleasant-day classification in Madrid.
- For Madrid-focused modeling, ClimateWins could prioritize temperature metrics and precipitation, supplemented by solar radiation, while deprioritizing pressure-based indicators without substantial loss in predictive accuracy.

Model Performance Conclusions

All-Stations Model (One Decade)

The all-stations model demonstrated high predictive accuracy and identified weather stations and feature groups that contributed most strongly to pleasant-day classification. This model successfully highlighted spatial differences across Europe and guided the selection of stations for deeper analysis.

Budapest Model

The Budapest-specific Random Forest achieved perfect test accuracy, indicating a strong relationship between local weather variables and pleasant-day outcomes. Feature importance results showed that precipitation and temperature were the dominant drivers.

Ljubljana Model

The Ljubljana model also achieved perfect test accuracy, reinforcing the presence of a clear predictive signal in the local weather data. Precipitation emerged as the most influential variable, followed by temperature and solar-related features, reflecting a climate regime where rainfall strongly shapes daily comfort.

Madrid Model

The Madrid model achieved near-perfect accuracy (99.98%), with temperature-related variables playing a more prominent role than precipitation. This highlights a distinct climatic pattern in which heat and thermal extremes are the primary determinants of pleasant weather, with rainfall acting as a secondary factor.

Recommendations for ClimateWins

Prioritize precipitation and temperature monitoring across regions

Across all models, precipitation and temperature variables consistently emerged as the strongest predictors of pleasant days, even though their relative importance varied by location. ClimateWins should prioritize maintaining and improving the accuracy, resolution, and reliability of precipitation and temperature sensors, as these variables deliver the greatest predictive value for pleasant-weather modeling.

Tailor modeling strategies to regional climate patterns

The city-specific models reveal that pleasant weather drivers differ by climate regime—with precipitation dominating in Budapest and Ljubljana, and temperature dominating in Madrid. ClimateWins should adopt region-specific modeling and investment strategies, rather than relying on a single universal set of indicators, to better capture local weather dynamics and improve forecast relevance.

Deprioritize low-impact variables to streamline data collection:

Variables such as atmospheric pressure, humidity, and cloud cover consistently showed low importance across models. ClimateWins could consider reducing emphasis on these indicators in pleasant-weather prediction workflows, allowing resources to be reallocated toward higher-impact measurements or advanced modeling techniques without substantially reducing predictive performance.

Final Reflections

Random Forest models are highly effective for structured climate data

The consistently high accuracy across all models demonstrates that Random Forests are well suited for identifying patterns in well-organized, multivariate weather datasets, particularly when relationships between features and outcomes are nonlinear.

Feature importance analysis adds practical value beyond prediction

Examining feature importance made it possible to move beyond accuracy metrics and identify which weather variables truly drive pleasant-day outcomes. This insight directly supports more informed decisions about data collection priorities and modeling focus.

Weather drivers vary meaningfully by location

While precipitation and temperature were important across all models, their relative influence differed by city. This reinforces the importance of regional context in climate modeling and cautions against one-size-fits-all approaches.

Not all data contributes equally to predictive performance

Several variables, such as pressure and humidity, consistently showed low importance. Recognizing which features add limited value is just as important as identifying strong predictors, particularly when resources are constrained.

High accuracy requires thoughtful interpretation

The near-perfect and perfect test accuracies observed in the city-specific models likely reflect strong, internally consistent patterns in the dataset rather than model misuse. However, these results highlight the importance of careful validation and awareness of potential overfitting when applying similar models to new locations, time periods, or operational forecasting contexts.

Applied judgment is as important as technical execution

This project underscored the role of analytical judgment, including how to structure datasets, interpret model outputs, and communicate results responsibly, as a critical skill in applying machine learning to real-world climate challenges.